

DAY 5

In []: `import pandas as pd`

In [4]: `df=pd.read_csv("cleaned file.csv")`

In [5]: `df`

Out[5]:

	Unnamed: 0	country_name	year	life_ladder	log_gdp_per_capita	social_support	health
0	0	Afghanistan	2008	3.724	7.370	0.451	
1	1	Afghanistan	2009	4.402	7.540	0.552	
2	2	Afghanistan	2010	4.758	7.647	0.539	
3	3	Afghanistan	2011	3.832	7.620	0.521	
4	4	Afghanistan	2012	3.783	7.705	0.521	
...	...	...	...	...	...	...	...
1920	1944	Zimbabwe	2016	3.735	7.984	0.768	
1921	1945	Zimbabwe	2017	3.638	8.016	0.754	
1922	1946	Zimbabwe	2018	3.616	8.049	0.775	
1923	1947	Zimbabwe	2019	2.694	7.950	0.759	
1924	1948	Zimbabwe	2020	3.160	7.829	0.717	

1925 rows × 12 columns



In [6]: `df.describe()`

Out[6]:

	Unnamed: 0	year	life_ladder	log_gdp_per_capita	social_support	health
count	1925.000000	1925.000000	1925.000000	1925.000000	1925.000000	
mean	973.437403	2013.231169	5.468297	9.366309	0.812817	
std	562.487417	4.162258	1.118544	1.143741	0.118451	
min	0.000000	2005.000000	2.375000	6.635000	0.290000	
25%	487.000000	2010.000000	4.640000	8.474000	0.750000	
50%	974.000000	2013.000000	5.387000	9.444000	0.836000	
75%	1462.000000	2017.000000	6.290000	10.335000	0.905000	
max	1948.000000	2020.000000	8.019000	11.648000	0.987000	



In [7]: `df.median(numeric_only=True)`

```
Out[7]: Unnamed: 0      974.000
        year      2013.000
        life_ladder      5.387
        log_gdp_per_capita      9.444
        social_support      0.836
        healthy_life_expectancy_at_birth      65.000
        freedom_to_make_life_choices      0.760
        generosity      -0.016
        perceptions_of_corruption      0.794
        positive_affect      0.722
        negative_affect      0.258
        dtype: float64
```

In [8]: `df["life_ladder"].quantile([0.25,0.5,0.75])`

```
Out[8]: 0.25    4.640
        0.50    5.387
        0.75    6.290
        Name: life_ladder, dtype: float64
```

In [9]: `df.quantile([0.25,0.5,0.75],numeric_only=True)`

```
Out[9]:
```

	Unnamed: 0	year	life_ladder	log_gdp_per_capita	social_support	healthy_life_exp
0.25	487.0	2010.0	4.640	8.474	0.750	
0.50	974.0	2013.0	5.387	9.444	0.836	
0.75	1462.0	2017.0	6.290	10.335	0.905	

◀  ▶

In [10]: `df.value_counts()`

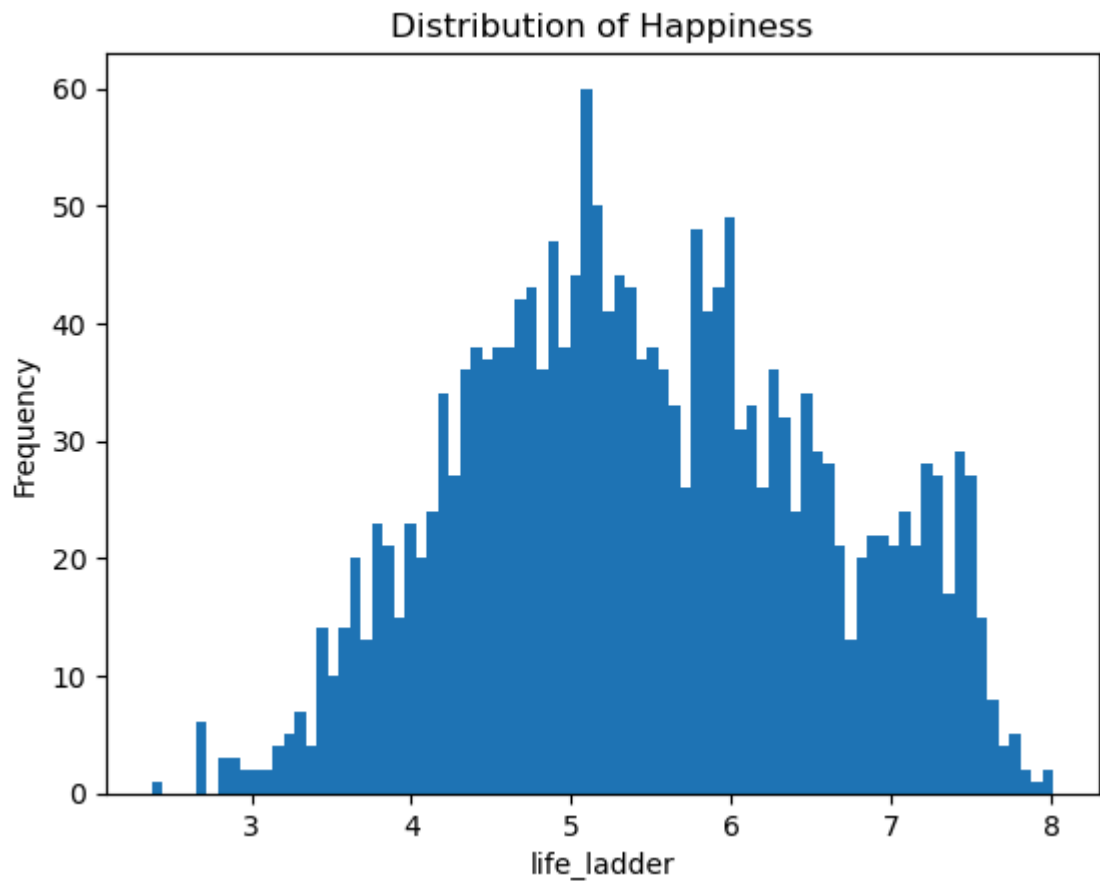
```
Out[10]: Unnamed: 0  country_name  year  life_ladder  log_gdp_per_capita  social_support
healthy_life_expectancy_at_birth  freedom_to_make_life_choices  generosity  per
ceptions_of_corruption  positive_affect  negative_affect
0      Afghanistan    2008    3.724      7.370      0.451
50.80      0.718      0.168      0.8
82      0.518      0.258      1
1218      Netherlands    2014    7.321      10.867      0.909
71.88      0.910      0.331      0.4
57      0.868      0.221      1
1306      Norway    2017    7.579      11.050      0.950
73.10      0.953      0.236      0.2
50      0.849      0.203      1
1305      Norway    2016    7.596      11.035      0.960
73.00      0.954      0.133      0.4
10      0.850      0.209      1
1304      Norway    2015    7.603      11.033      0.947
72.90      0.948      0.257      0.2
99      0.843      0.209      1

..
643      Greece    2014    4.756      10.257      0.832
71.74      0.369      -0.288      0.9
30      0.695      0.385      1
642      Greece    2013    4.720      10.243      0.687
71.68      0.426      -0.272      0.9
41      0.689      0.482      1
641      Greece    2012    5.096      10.268      0.812
71.62      0.373      -0.305      0.9
59      0.581      0.352      1
640      Greece    2011    5.372      10.339      0.852
71.56      0.528      -0.316      0.9
41      0.591      0.323      1
1948      Zimbabwe    2020    3.160      7.829      0.717
56.80      0.643      -0.009      0.7
89      0.703      0.346      1
Name: count, Length: 1925, dtype: int64
```

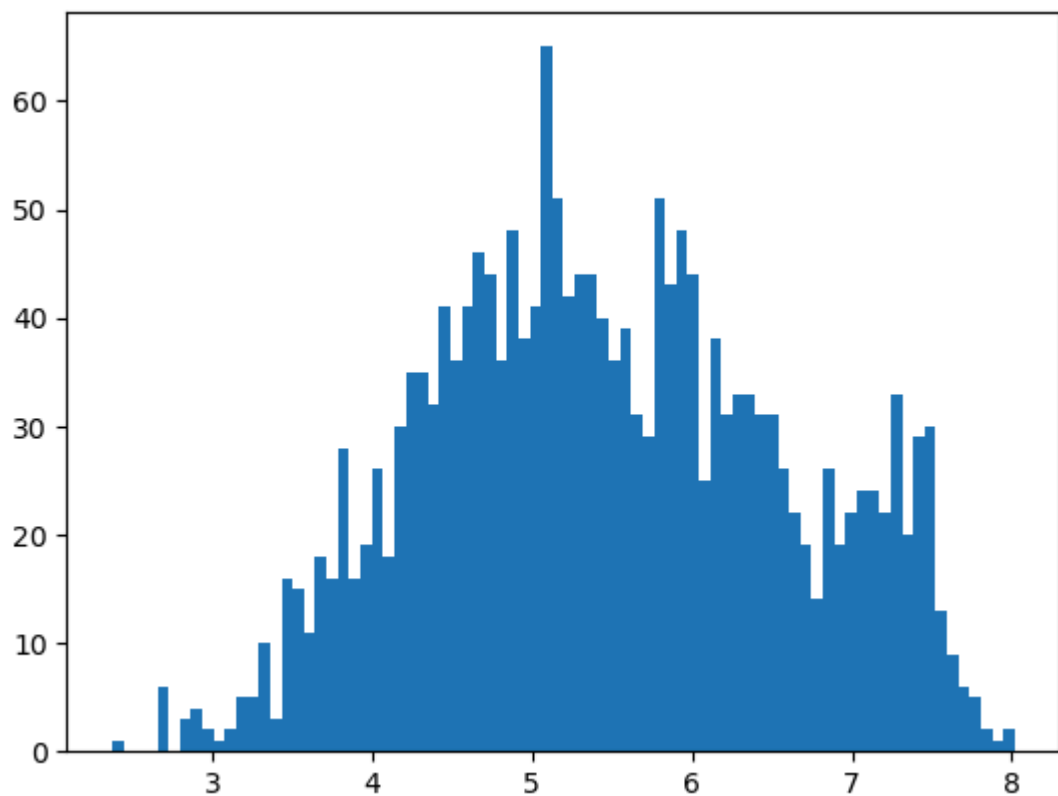
NORMAL DISTRIBUTION

```
In [11]: import matplotlib.pyplot as plt

plt.hist(df["life_ladder"], bins=82)
plt.xlabel("life_ladder")
plt.ylabel("Frequency")
plt.title("Distribution of Happiness")
plt.show()
```



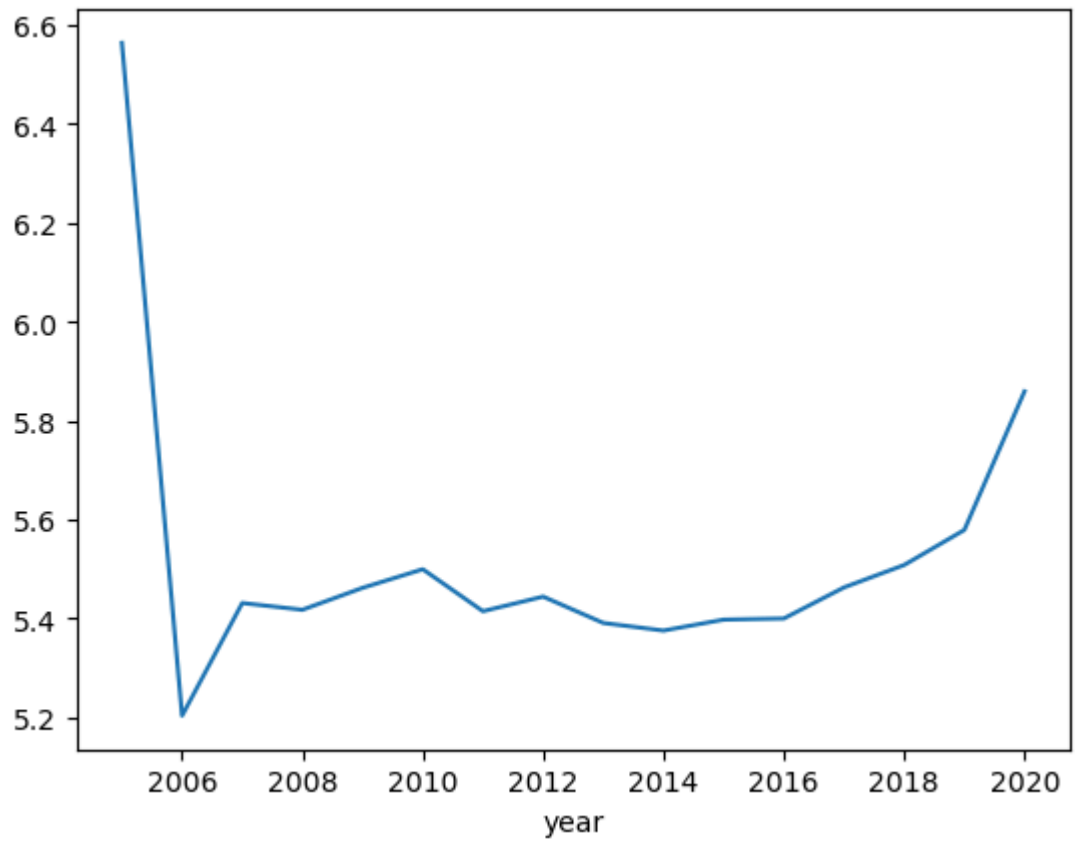
```
In [14]: plt.hist(df["life_ladder"], bins=80)
plt.show()
```



HAPPINES OVER YEARS

```
In [12]: df.groupby("year")["life_ladder"].mean().plot()
```

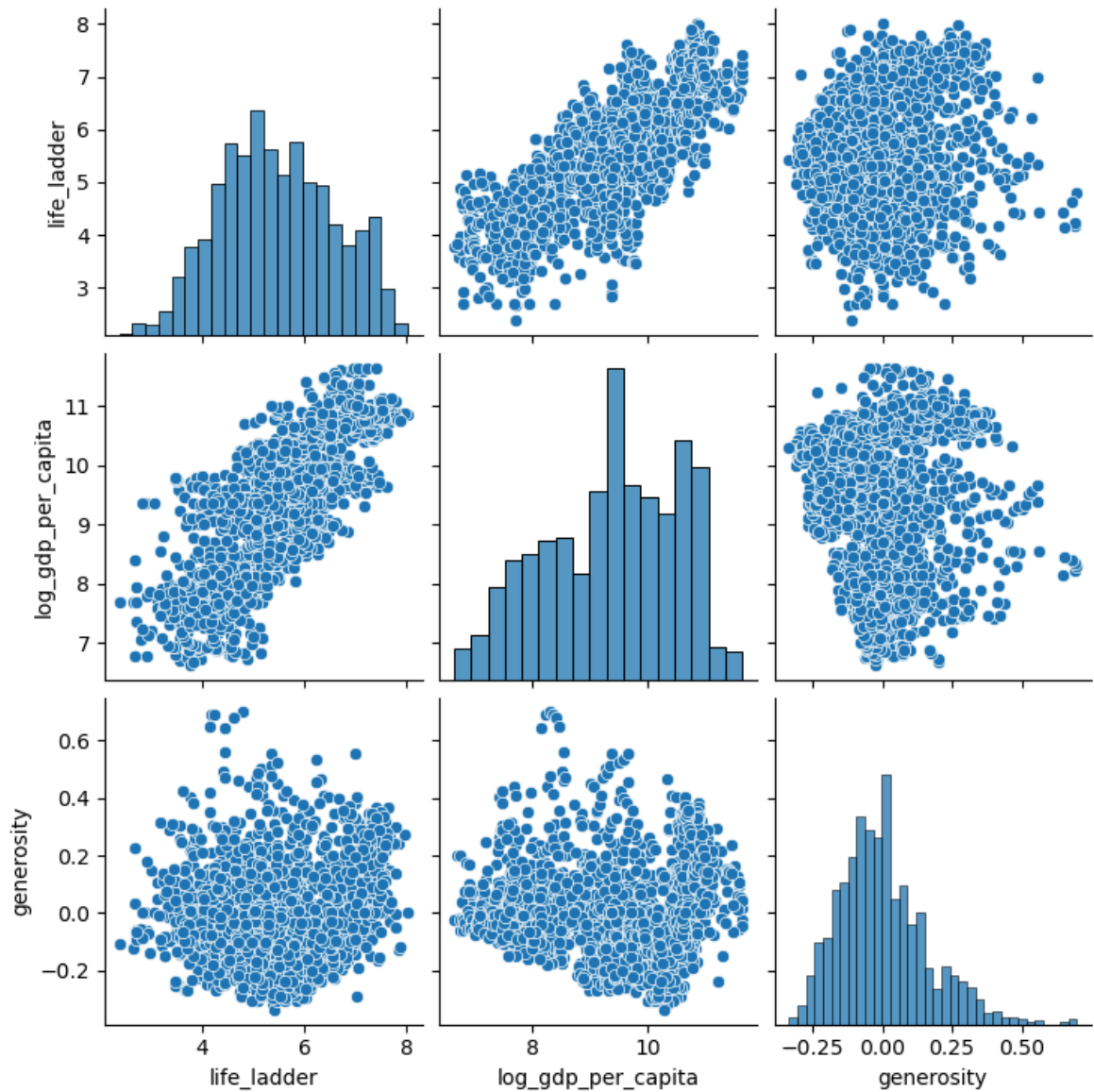
Out[12]: <Axes: xlabel='year'>



RELATION

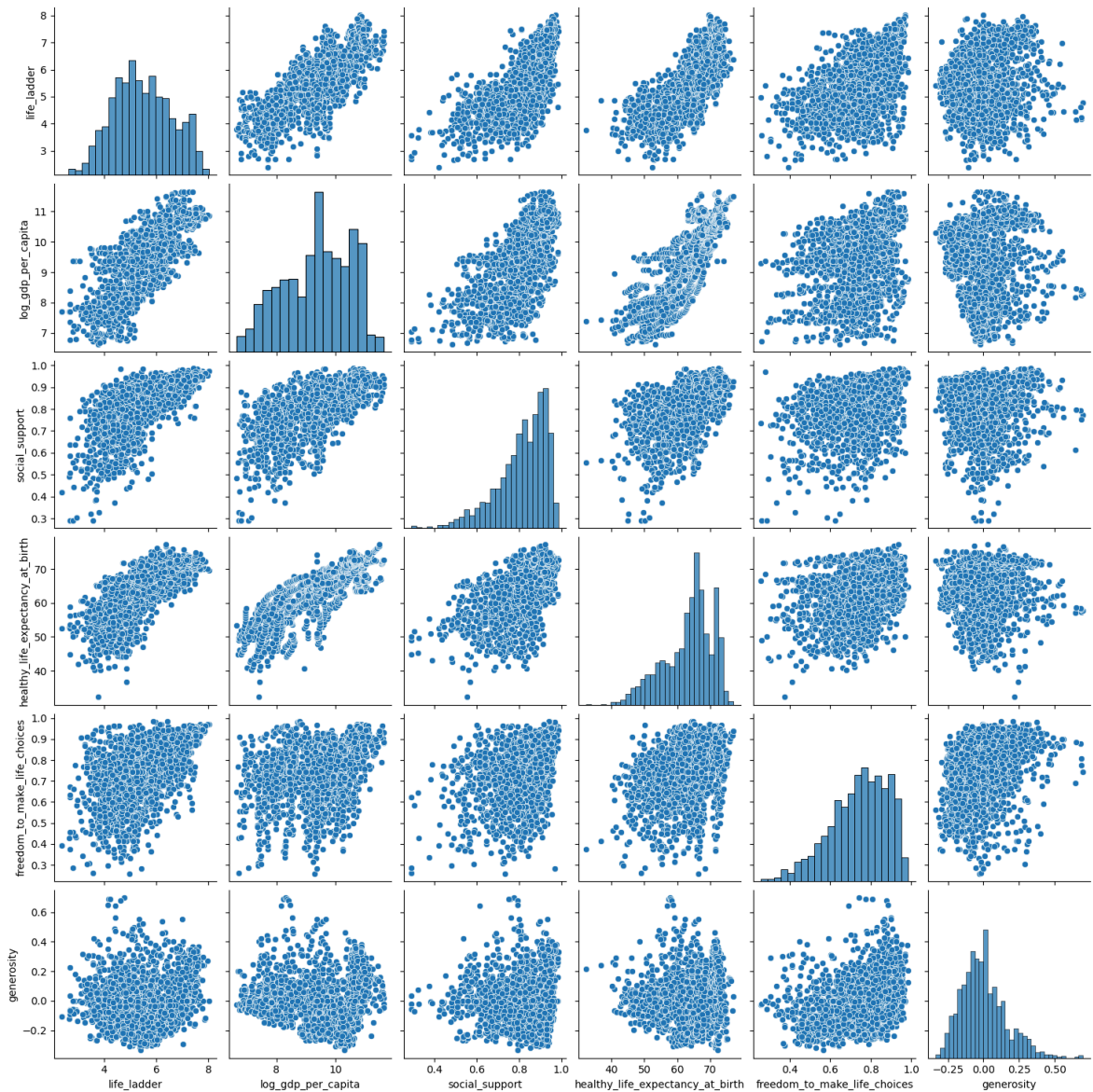
```
In [16]: import seaborn as sns
sns.pairplot(df[["life_ladder", "log_gdp_per_capita", "generosity"]])
```

Out[16]: <seaborn.axisgrid.PairGrid at 0x2e586667cb0>



```
In [18]: sns.pairplot(df[[
    "life_ladder",
    "log_gdp_per_capita",
    "social_support",
    "healthy_life_expectancy_at_birth",
    "freedom_to_make_life_choices",
    "generosity"
]])
```

Out[18]: <seaborn.axisgrid.PairGrid at 0x2e587ee7ed0>



PIE CHART

```
In [17]: import matplotlib.pyplot as plt
import pandas as pd

df["happiness_level"] = pd.cut(
    df["life_ladder"],
    bins=[0, 4, 6, 10],
    labels=["Low", "avarage", "High"]
)

counts = df["happiness_level"].value_counts()

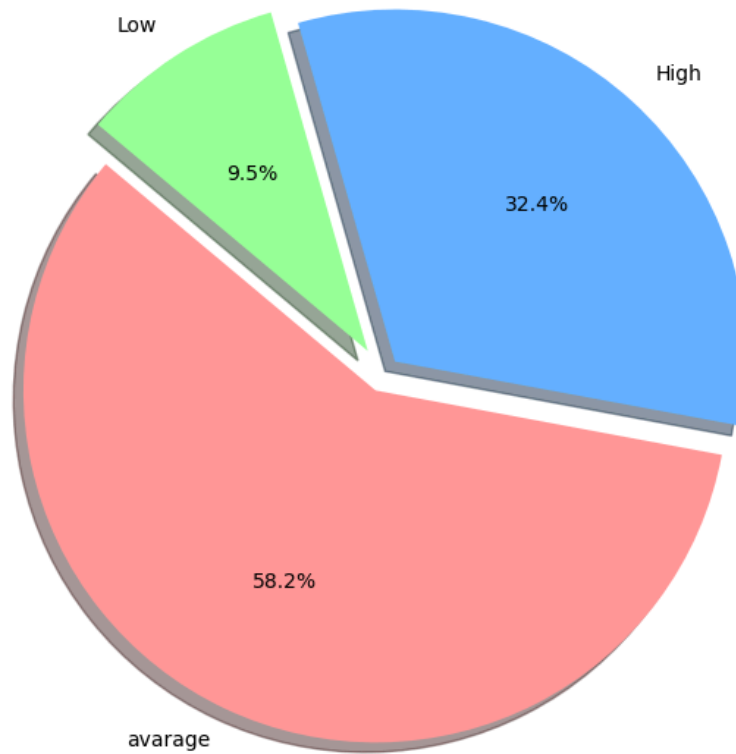
plt.figure(figsize=(7,7))

colors = ["#ff9999", "#66b3ff", "#99ff99"]
explode = (0.05, 0.05, 0.08)

plt.pie(
    counts,
    labels=counts.index,
    autopct="%1.1f%%",
    startangle=500,
    colors=colors,
```

```
explode=explode,  
shadow=True  
)  
  
plt.title("Distribution of Happiness Levels", fontsize=35)  
plt.axis("equal")  
  
plt.show()
```

Distribution of Happiness Levels



In [19]: df

Out[19]:

	Unnamed: 0	country_name	year	life_ladder	log_gdp_per_capita	social_support	healthy_life_exp
0	0	Afghanistan	2008	3.724	7.370	0.451	
1	1	Afghanistan	2009	4.402	7.540	0.552	
2	2	Afghanistan	2010	4.758	7.647	0.539	
3	3	Afghanistan	2011	3.832	7.620	0.521	
4	4	Afghanistan	2012	3.783	7.705	0.521	
...	...	...	...	...	...	...	...
1920	1944	Zimbabwe	2016	3.735	7.984	0.768	
1921	1945	Zimbabwe	2017	3.638	8.016	0.754	
1922	1946	Zimbabwe	2018	3.616	8.049	0.775	
1923	1947	Zimbabwe	2019	2.694	7.950	0.759	
1924	1948	Zimbabwe	2020	3.160	7.829	0.717	

1925 rows × 13 columns

In [20]: `df = df.drop("Unnamed: 0", axis=1)`In [21]: `df`

Out[21]:

	country_name	year	life_ladder	log_gdp_per_capita	social_support	healthy_life_exp
0	Afghanistan	2008	3.724	7.370	0.451	
1	Afghanistan	2009	4.402	7.540	0.552	
2	Afghanistan	2010	4.758	7.647	0.539	
3	Afghanistan	2011	3.832	7.620	0.521	
4	Afghanistan	2012	3.783	7.705	0.521	
...	...	...	...	...	...	...
1920	Zimbabwe	2016	3.735	7.984	0.768	
1921	Zimbabwe	2017	3.638	8.016	0.754	
1922	Zimbabwe	2018	3.616	8.049	0.775	
1923	Zimbabwe	2019	2.694	7.950	0.759	
1924	Zimbabwe	2020	3.160	7.829	0.717	

1925 rows × 12 columns

In [22]: `df`

Out[22]:

	country_name	year	life_ladder	log_gdp_per_capita	social_support	healthy_life_e
0	Afghanistan	2008	3.724	7.370	0.451	
1	Afghanistan	2009	4.402	7.540	0.552	
2	Afghanistan	2010	4.758	7.647	0.539	
3	Afghanistan	2011	3.832	7.620	0.521	
4	Afghanistan	2012	3.783	7.705	0.521	
...	...	...	...	...	...	...
1920	Zimbabwe	2016	3.735	7.984	0.768	
1921	Zimbabwe	2017	3.638	8.016	0.754	
1922	Zimbabwe	2018	3.616	8.049	0.775	
1923	Zimbabwe	2019	2.694	7.950	0.759	
1924	Zimbabwe	2020	3.160	7.829	0.717	

1925 rows × 12 columns



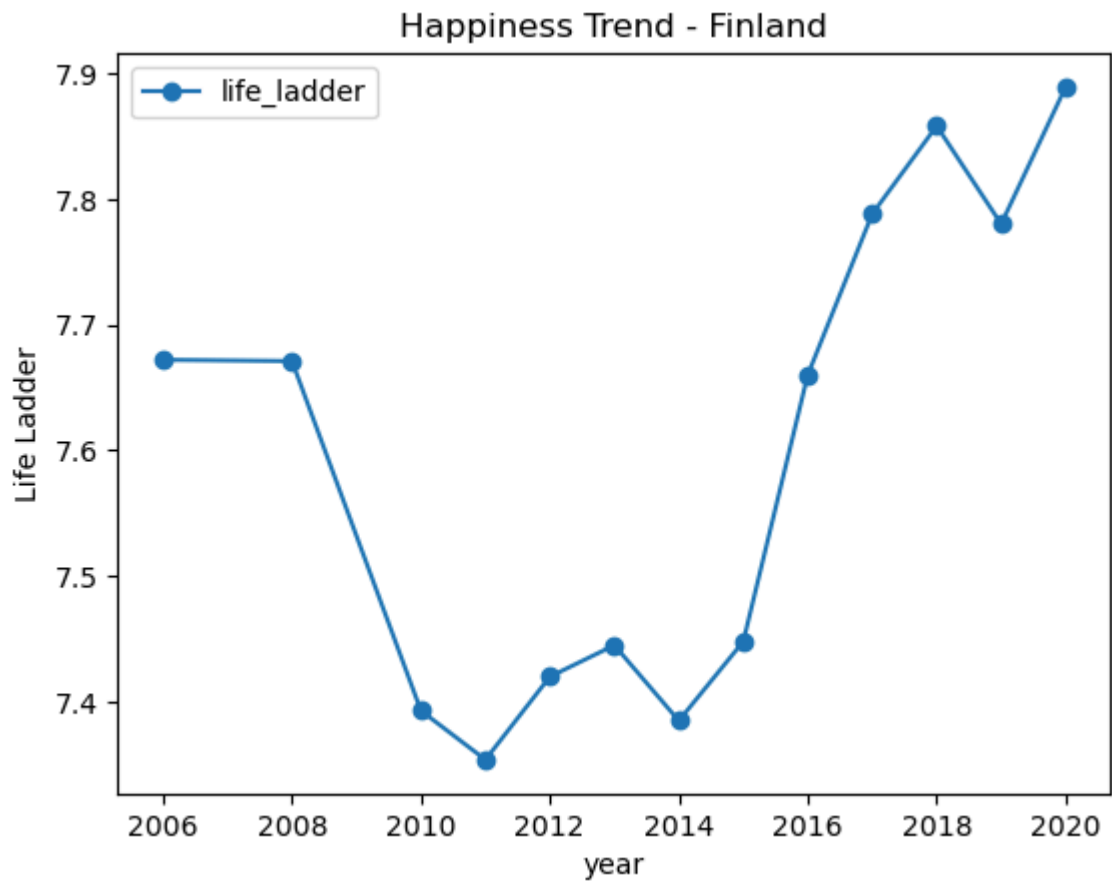
DAY 6

In [23]:

```
country_data = df[df["country_name"] == "Finland"]

country_data.sort_values("year").plot(
    x="year", y="life_ladder", kind="line", marker="o"
)

import matplotlib.pyplot as plt
plt.title("Happiness Trend - Finland")
plt.ylabel("Life Ladder")
plt.show()
```



The happiness trend in finland over the years

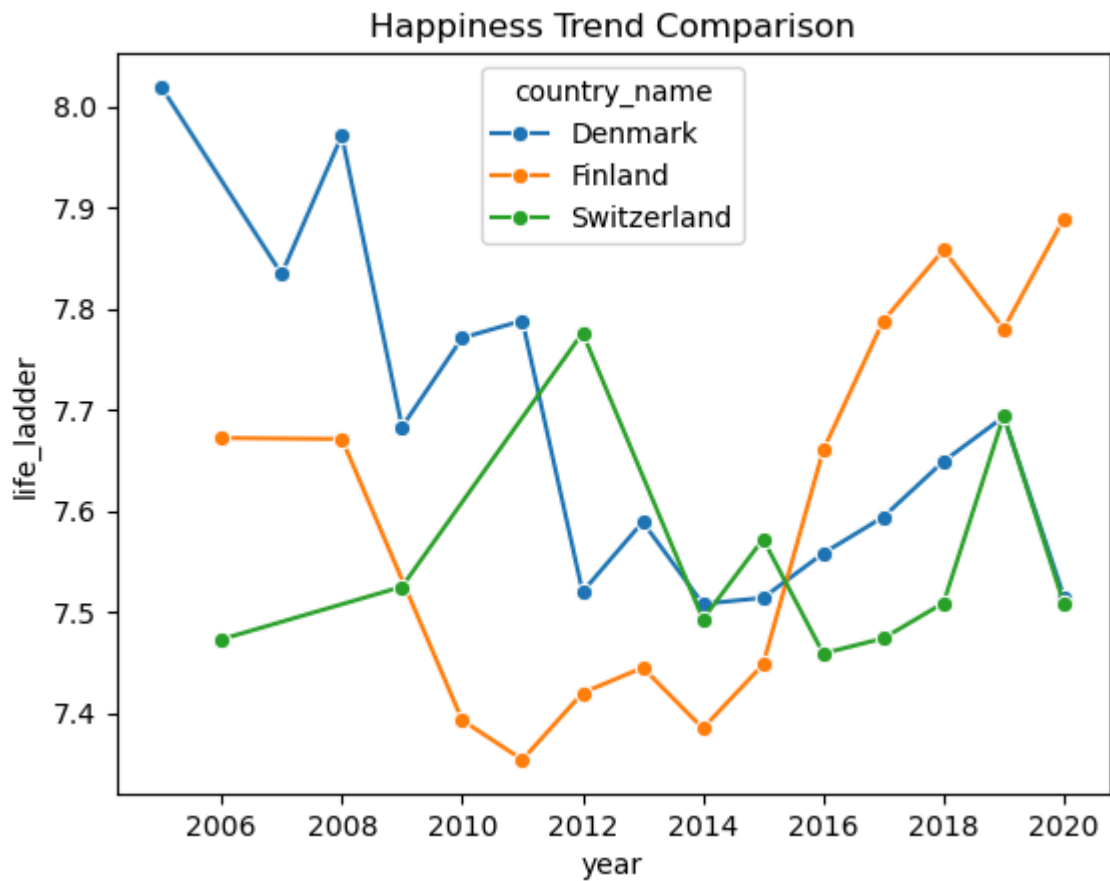
```
In [41]: countries = ["Finland", "Denmark", "Switzerland"]

subset = df[df["country_name"].isin(countries)]

import seaborn as sns

sns.lineplot(
    data=subset,
    x="year",
    y="life_ladder",
    hue="country_name",
    marker="o"
)

plt.title("Happiness Trend Comparison")
plt.show()
```

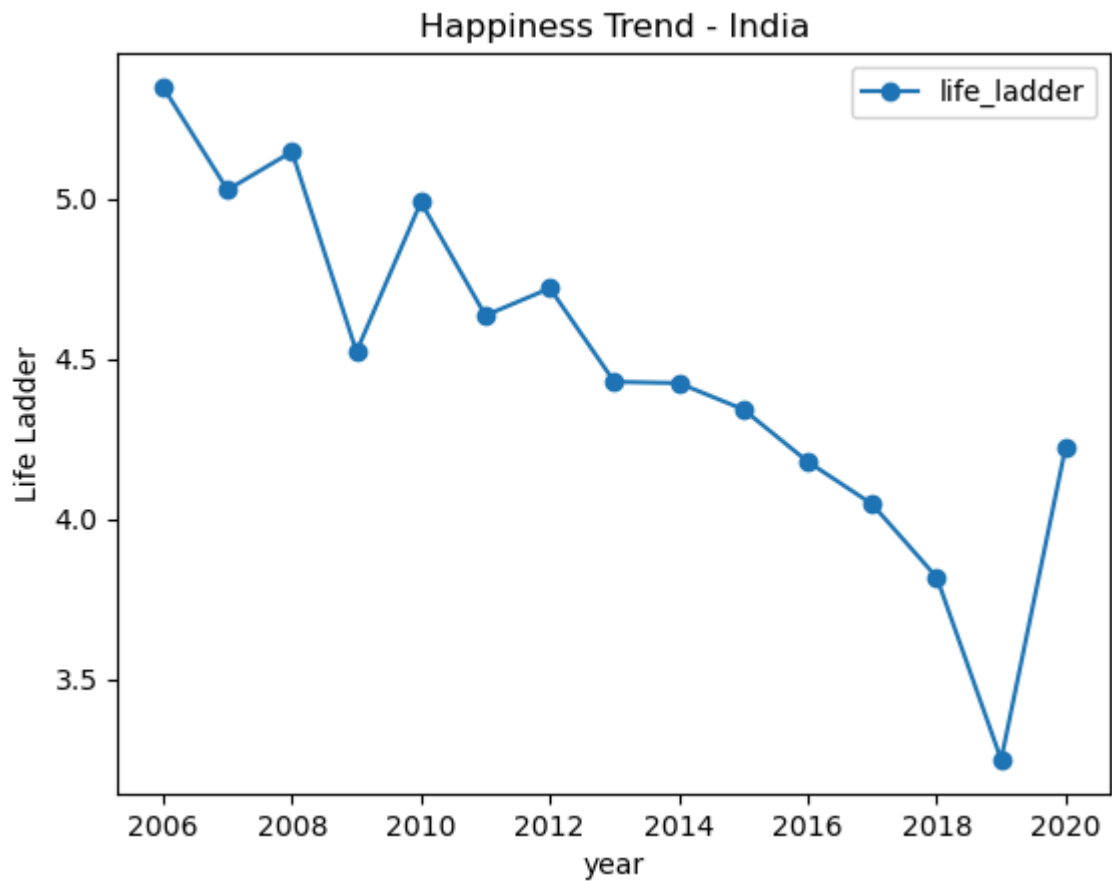


The happiness trend in specific countries over the years .The happiness trend in denmark is decreasing over the years .The happiness trend in finland is increasing over the years

```
In [25]: country_data = df[df["country_name"] == "India"]

country_data.sort_values("year").plot(
    x="year", y="life_ladder", kind="line", marker="o"
)

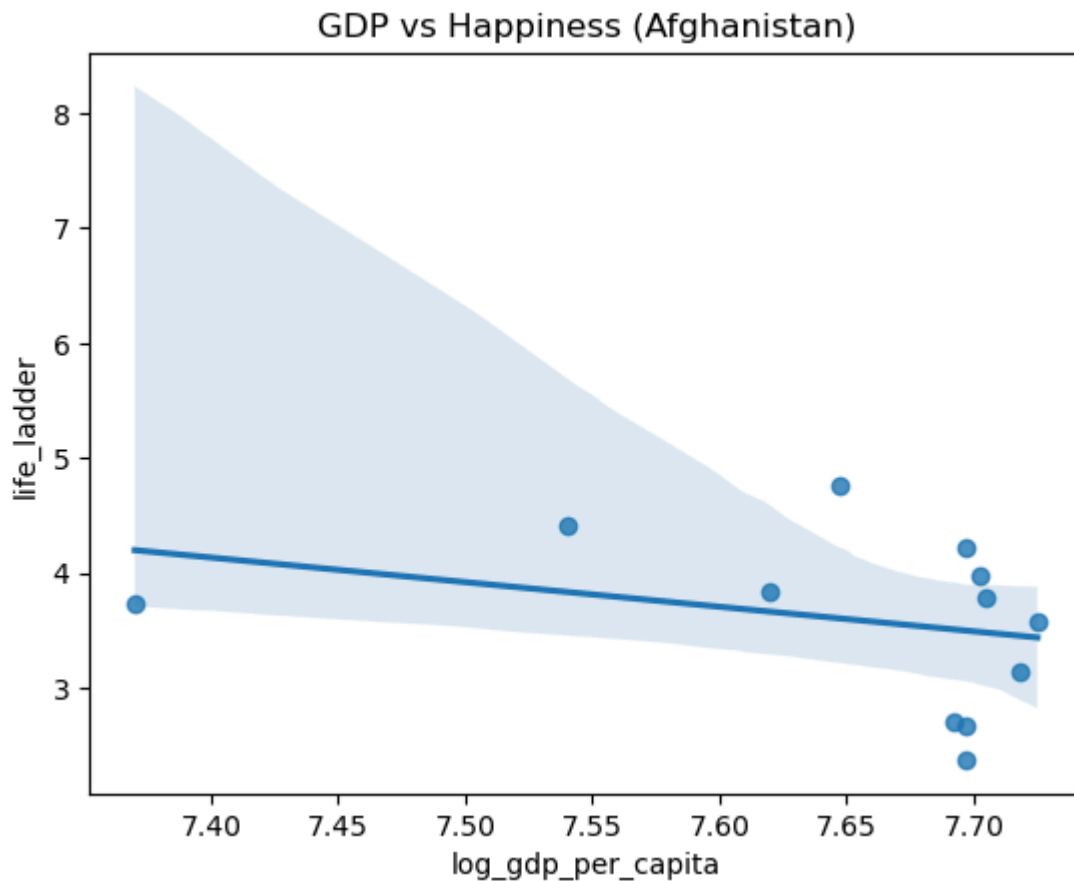
import matplotlib.pyplot as plt
plt.title("Happiness Trend - India")
plt.ylabel("Life Ladder")
plt.show()
```



Happiness trent in india over the years

```
In [40]: afg_df = df[df["country_name"] == "Afghanistan"]
```

```
In [28]: sns.regplot(  
    data=afg_df,  
    x="log_gdp_per_capita",  
    y="life_ladder"  
)  
  
plt.title("GDP vs Happiness (Afghanistan)")  
plt.show()
```



relation between gdp and happiness

```
In [33]: country_mean = df.groupby("country_name")["life_ladder"].mean()

rank = country_mean.rank(ascending=False)

india_rank = rank["India"]

print("India's Rank:", int(india_rank))
```

India's Rank: 126

```
In [34]: country_mean
```

```
Out[34]: country_name
Afghanistan    3.594667
Albania        5.019385
Algeria        5.379286
Angola         4.420250
Argentina      6.310133
...
Venezuela     6.019867
Vietnam       5.315923
Yemen         3.912250
Zambia        4.551714
Zimbabwe      3.882600
Name: life_ladder, Length: 164, dtype: float64
```

```
In [42]: df_2020 = df[df['year'] == 2020]

df_2020 = df_2020.sort_values(by='life_ladder', ascending=False)
```

```
df_2020['rank'] = range(1, len(df_2020) + 1)

argentina_rank = df_2020[df_2020['country_name'] == 'Argentina']

print(argentina_rank[['country_name', 'life_ladder', 'rank']])
```

```
country_name  life_ladder  rank
50    Argentina      5.901    47
```

In [36]: df

Out[36]:

	country_name	year	life_ladder	log_gdp_per_capita	social_support	healthy_life_e
0	Afghanistan	2008	3.724	7.370	0.451	
1	Afghanistan	2009	4.402	7.540	0.552	
2	Afghanistan	2010	4.758	7.647	0.539	
3	Afghanistan	2011	3.832	7.620	0.521	
4	Afghanistan	2012	3.783	7.705	0.521	
...	...	...	...	...	...	...
1920	Zimbabwe	2016	3.735	7.984	0.768	
1921	Zimbabwe	2017	3.638	8.016	0.754	
1922	Zimbabwe	2018	3.616	8.049	0.775	
1923	Zimbabwe	2019	2.694	7.950	0.759	
1924	Zimbabwe	2020	3.160	7.829	0.717	

1925 rows × 12 columns

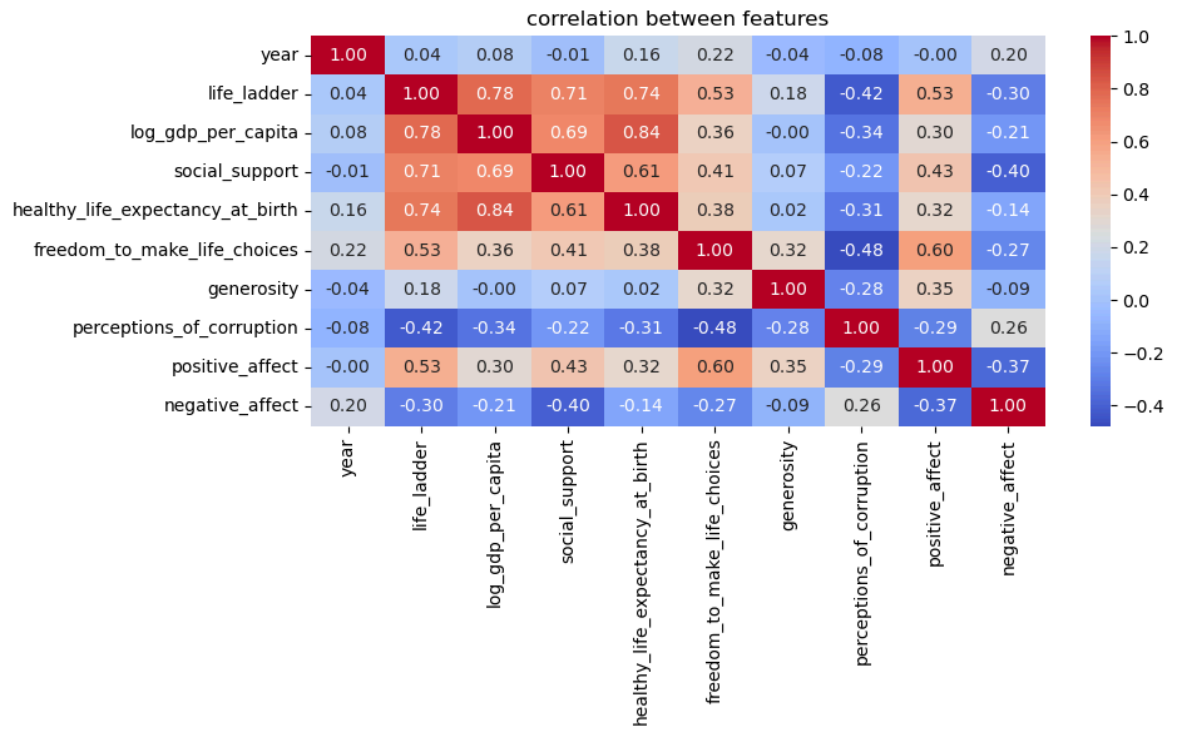


CORRELATION

```
In [ ]: plt.figure(figsize=(10,6))

sns.heatmap(df.corr(numeric_only=True),annot=True,fmt=".2f",cmap="coolwarm")

plt.title("correlation between features")
plt.tight_layout()
plt.show()
\
```



In [44]: df

Out[44]:

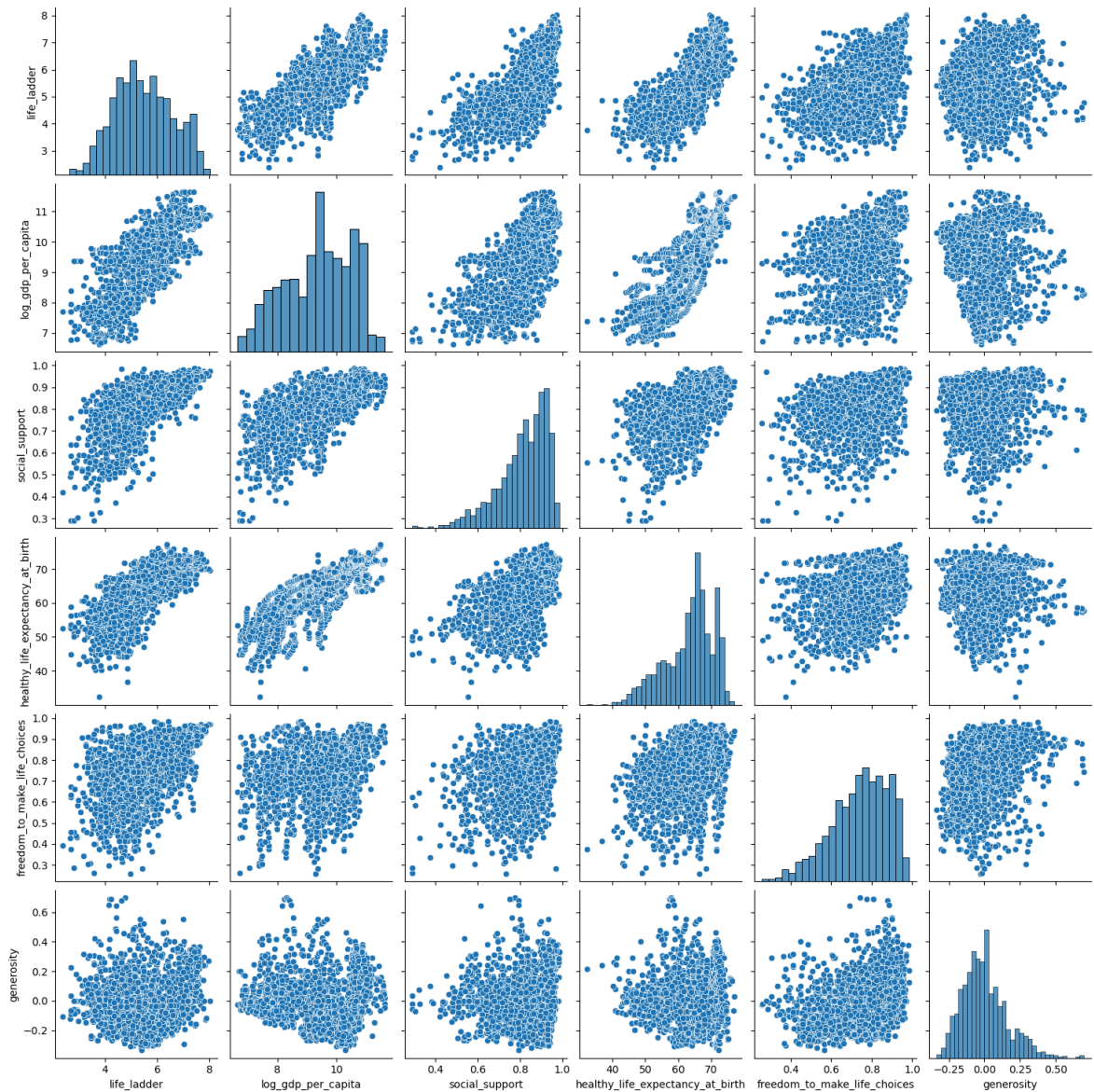
	country_name	year	life_ladder	log_gdp_per_capita	social_support	healthy_life_
0	Afghanistan	2008	3.724	7.370	0.451	
1	Afghanistan	2009	4.402	7.540	0.552	
2	Afghanistan	2010	4.758	7.647	0.539	
3	Afghanistan	2011	3.832	7.620	0.521	
4	Afghanistan	2012	3.783	7.705	0.521	
...	...	...	...	...	...	...
1920	Zimbabwe	2016	3.735	7.984	0.768	
1921	Zimbabwe	2017	3.638	8.016	0.754	
1922	Zimbabwe	2018	3.616	8.049	0.775	
1923	Zimbabwe	2019	2.694	7.950	0.759	
1924	Zimbabwe	2020	3.160	7.829	0.717	

1925 rows × 12 columns

In [46]:

```
sns.pairplot(df[[
    "life_ladder",
    "log_gdp_per_capita",
    "social_support",
    "healthy_life_expectancy_at_birth",
    "freedom_to_make_life_choices",
    "generosity"]])
```

Out[46]: <seaborn.axisgrid.PairGrid at 0x2e58b42d810>



```
In [47]: import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))

sns.regplot(
    data=df,
    x="healthy_life_expectancy_at_birth",
    y="life_ladder"
)

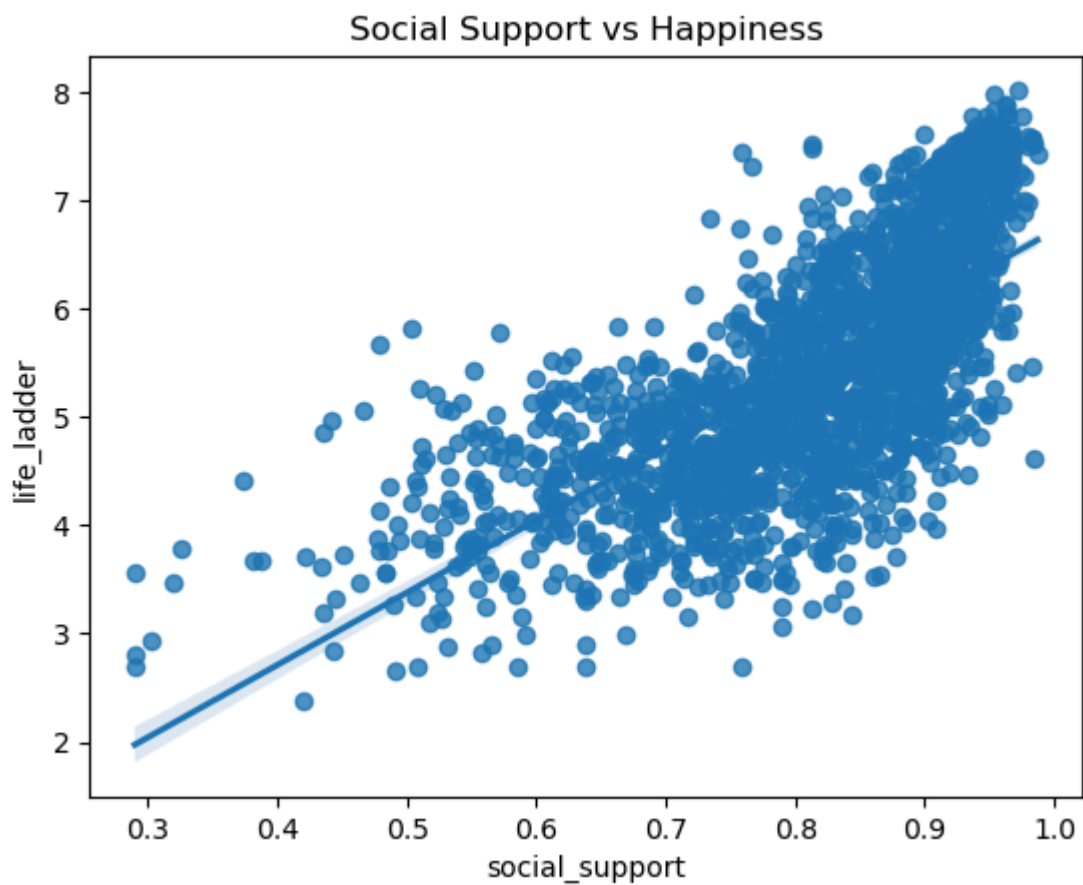
plt.title("Happiness vs Health")
plt.xlabel("Healthy Life Expectancy")
plt.ylabel("Life Ladder (Happiness)")

plt.tight_layout()
plt.show()
```



happines and health is directly propotional

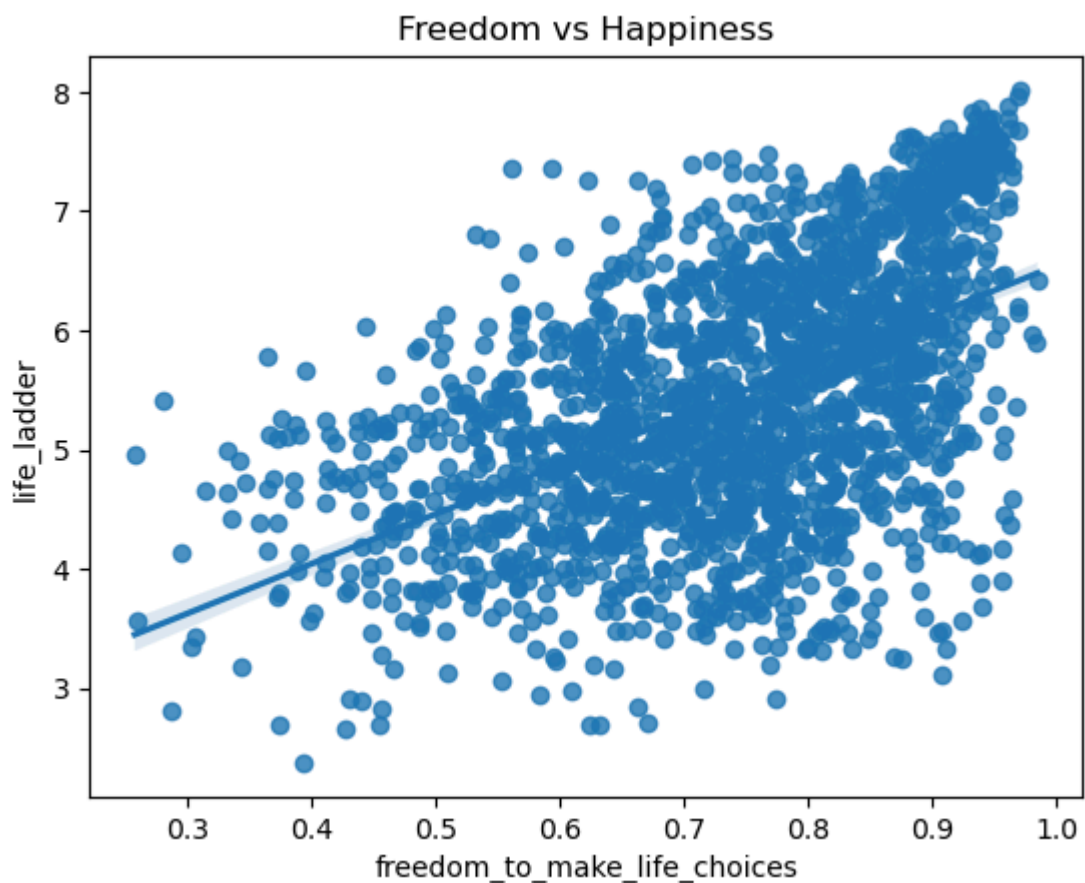
```
In [48]: sns.regplot(data=df, x="social_support", y="life_ladder")  
plt.title("Social Support vs Happiness")  
plt.show()
```



```
In [49]: sns.scatterplot(data=df, x="generosity", y="life_ladder")  
plt.title("Generosity vs Happiness")  
plt.show()
```

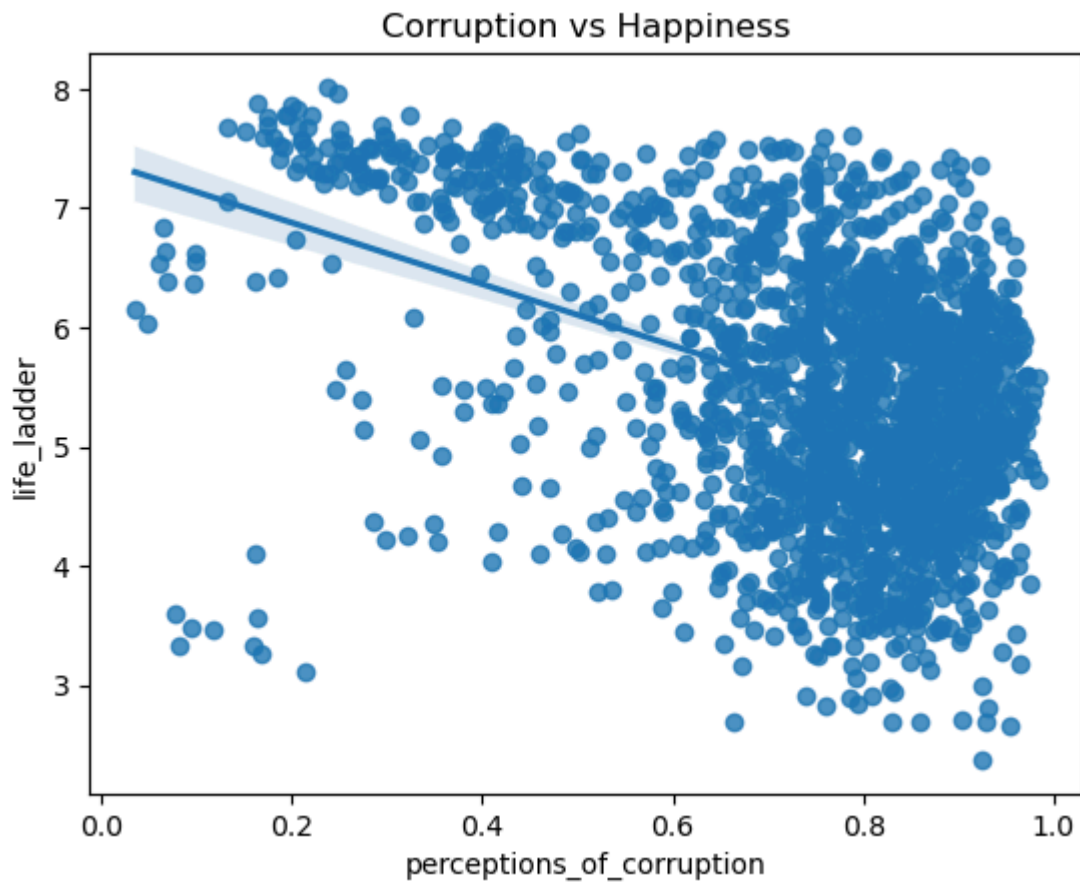


```
In [50]: sns.regplot(data=df, x="freedom_to_make_life_choices", y="life_ladder")  
plt.title("Freedom vs Happiness")  
plt.show()
```



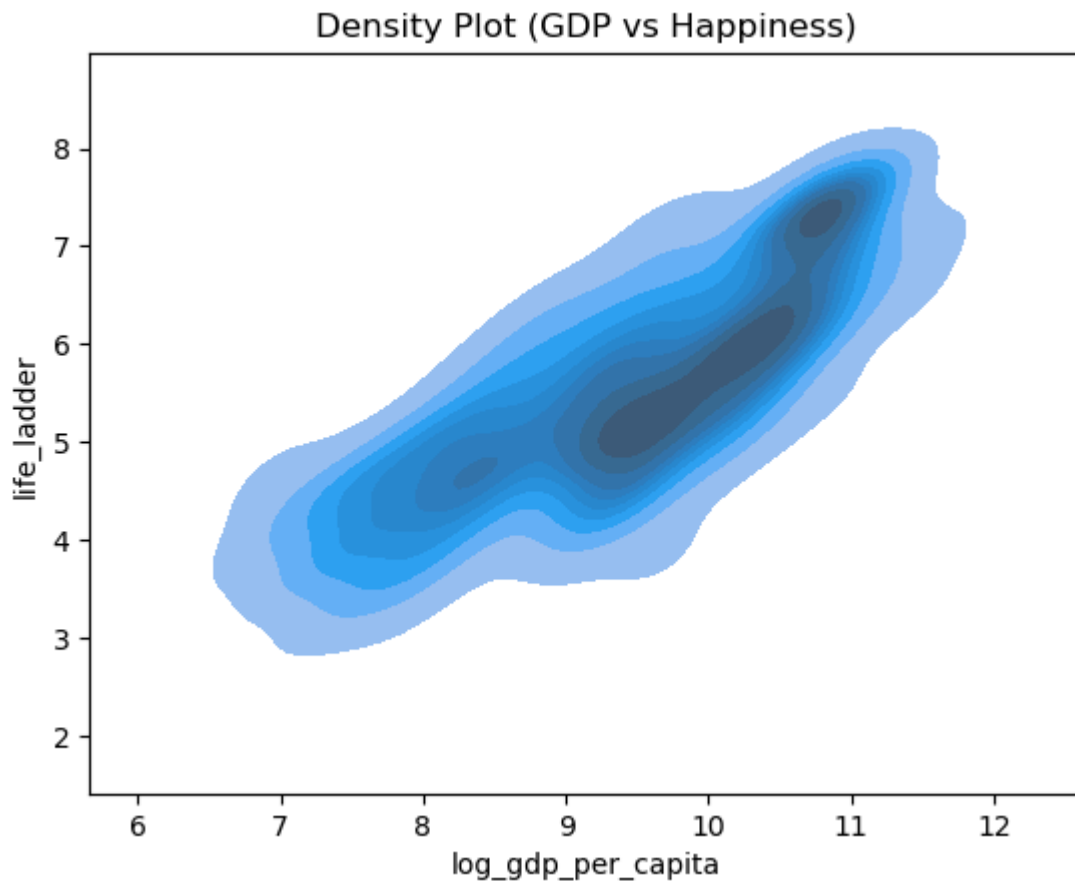
freedom and happines is directly propotional

```
In [51]: sns.regplot(data=df, x="perceptions_of_corruption", y="life_ladder")  
plt.title("Corruption vs Happiness")  
plt.show()
```



corruption and happiness is indirectly proportional

```
In [52]: sns.kdeplot(  
    data=df,  
    x="log_gdp_per_capita",  
    y="life_ladder",  
    fill=True  
)  
  
plt.title("Density Plot (GDP vs Happiness)")  
plt.show()
```

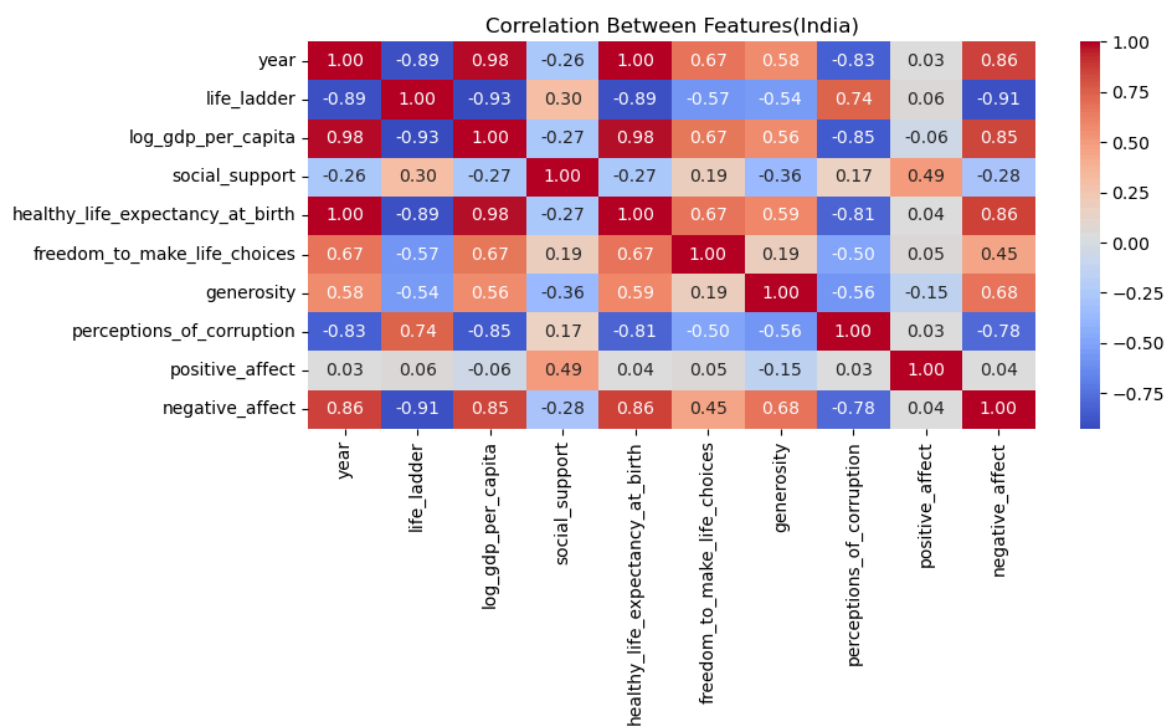


```
In [55]: india_dt= df[df["country_name"]=="India"]
```

```
In [56]: plt.figure(figsize=(10,6))

sns.heatmap(india_dt.corr(numeric_only=True), annot=True, fmt=".2f", cmap="coolw

plt.title("Correlation Between Features(India)")
plt.tight_layout()
plt.show()
```



correlation between features in india

top 10 rich country

```
In [57]: top10 = (
df.groupby("country_name")["log_gdp_per_capita"]
    .mean()
    .sort_values(ascending=False)
    .head(10)
)

print(top10)
```

country_name	
Luxembourg	11.607091
Qatar	11.510667
Singapore	11.329000
Switzerland	11.095400
United Arab Emirates	11.066727
Ireland	11.064857
Norway	11.039100
Kuwait	10.977800
United States	10.956067
Austria	10.886846

Name: log_gdp_per_capita, dtype: float64

```
In [58]: df_2020 = df[df['year'] == 2020]

df_2020 = df_2020.sort_values(by='life_ladder', ascending=False)

df_2020['rank'] = range(1, len(df_2020) + 1)

argentina_rank = df_2020[df_2020['country_name'] == 'Argentina']

print(argentina_rank[['country_name', 'life_ladder', 'rank']])
```

	country_name	life_ladder	rank
50	Argentina	5.901	47

```
In [60]: country_mean = df.groupby("country_name")["life_ladder"].mean()

rank = country_mean.rank(ascending=False)

india_rank = rank["India"]

print("India's Rank:", int(india_rank))
```

India's Rank: 126

```
In [59]: country_mean = df.groupby("country_name")["life_ladder"].mean()

last_country = country_mean.idxmin()

print("Last Country:", last_country)
```

Last Country: South Sudan

```
In [61]: countries = ["Afghanistan", "India", "Finland", "United States"]
```



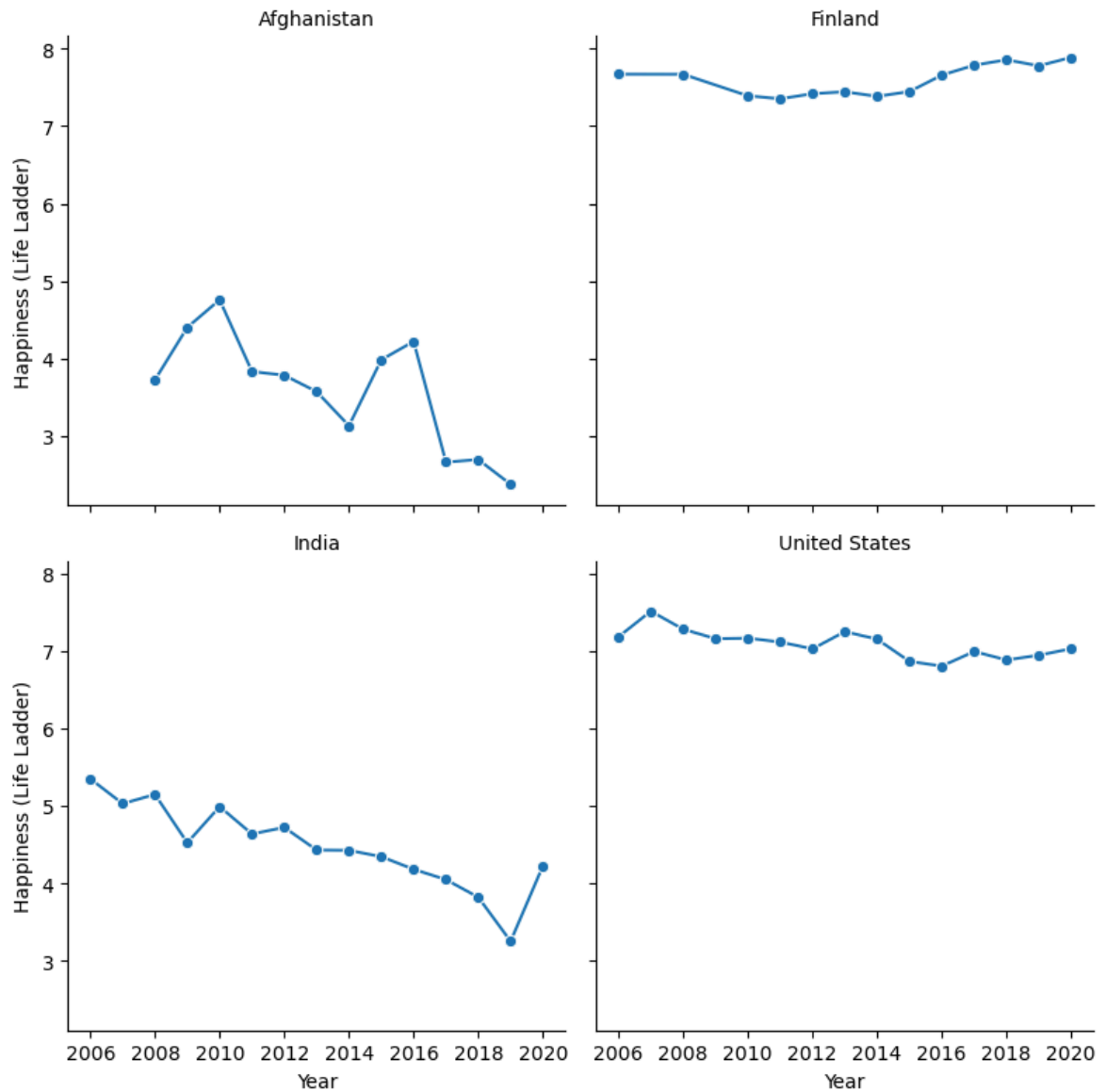
```
subset = df[df["country_name"].isin(countries)]

g = sns.FacetGrid(subset, col="country_name", col_wrap=2, height=4)

g.map(sns.lineplot, "year", "life_ladder", marker="o")

g.set_titles("{col_name}")
g.set_axis_labels("Year", "Happiness (Life Ladder)")

plt.tight_layout()
plt.show()
```



In []: